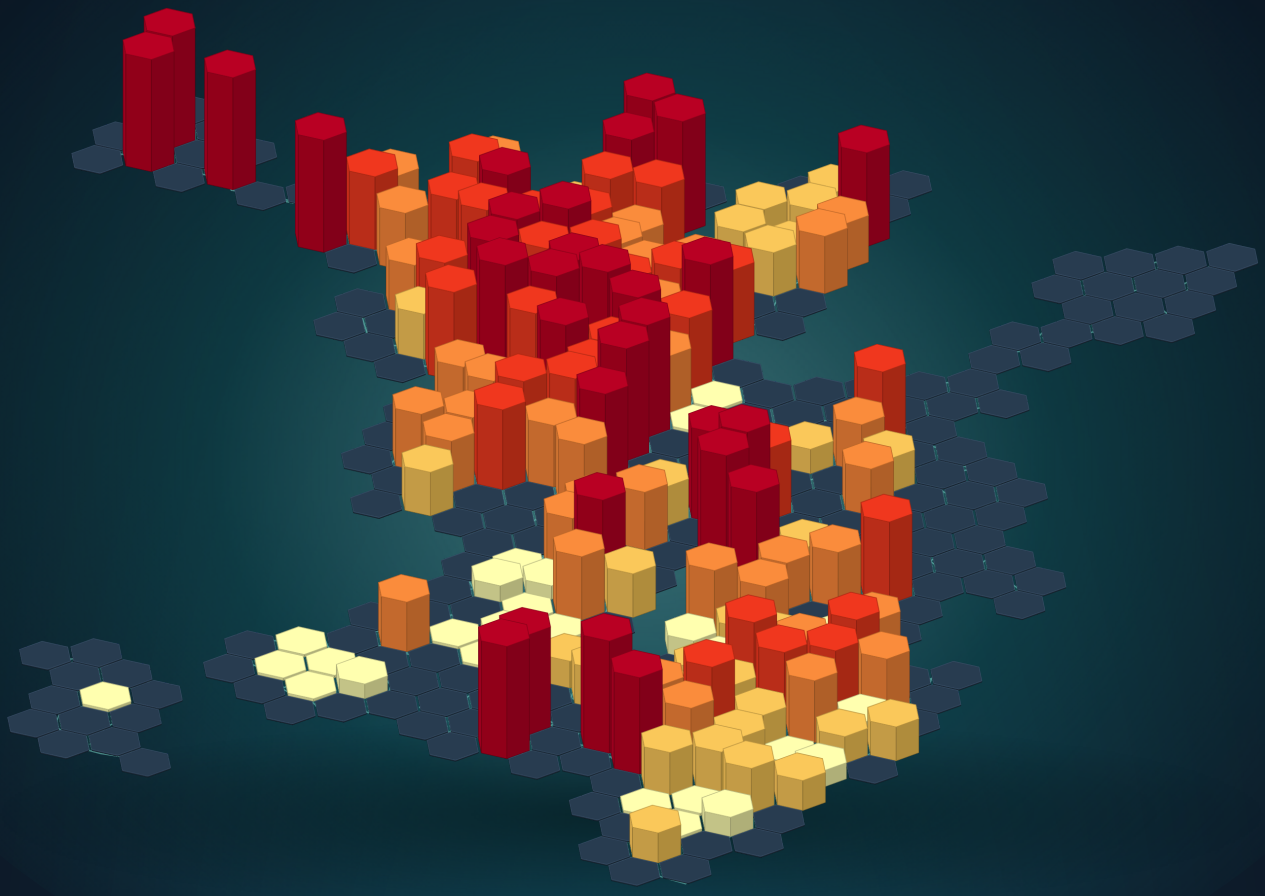
GIS PORTFOLIO · DATA GOVERNANCE

Geospatial Data Quality & Governance Framework

A rule-based quality assessment of OpenStreetMap buildings, roads and points of interest for the Dammam–Khobar urban area.



110,818	15	8 / 13	63.3%
features assessed	quality rules · 6 dimensions	compliance targets met	major roads named · target 95%

Share of unnamed roads per 1 km² hexagon · bar height = share

Norah Alhumud

Data Quality & Analysis · Geospatial Information

November 2026 · Data © OpenStreetMap contributors (ODbL)

01 · AT A GLANCE

Structurally sound, incomplete in attributes

Identifiers, geometry and coded values pass almost everywhere. The weak point is completeness: only **63.3% of major roads** carry a name or route number, against a 95% target.

110,818
Buildings, road segments and POIs checked

8 of 13
Rules meeting their compliance target

63.3%
Major roads named (target 95%)

103
Unnamed major roads with no named neighbour

Key findings

1. **No duplicate IDs, valid road codes.** No duplicate IDs were detected across the three layers. Road code-validity checks passed, while one building failed geometry validation.
2. **Names are the gap.** 252 of 686 major roads and 518 of 705 tertiary roads have no name or route number.
3. **The gap is concentrated.** Central and northern Dammam score worst; 72 of 170 rated 1 km² cells have more than half their roads unnamed.

Scope

Source	OpenStreetMap · Geofabrik, 16-09-2026
Licence	Open Database License
Buildings	91,212 polygons
Roads	18,403 segments
POIs	1,203 points
Study area	Dammam + Khobar, 206.4 km²
CRS	WGS 84 · UTM zone 39N
Tools	ArcGIS Pro · ArcPy

STEP 1
Data
Open OSM extract, urban area derived from building density

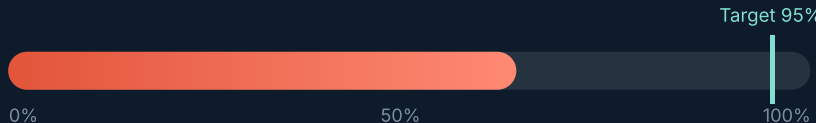
STEP 2
Rules
15 rules, severity and target for each

STEP 3
Tests
Spatial and attribute checks in ArcPy

STEP 4
Classify
Real error, optional field or needs source check

STEP 5
Indicators
Compliance, maps and interactive dashboard

63.3%
of major roads carry a name or route number (434 of 686)



15 rules, measured against targets

Each rule tests one dimension and carries a severity: **Error** blocks use, Warning lowers value, Review needs a source check. Targets are set by this project for assessment, based on how each attribute is used; they are not official OSM or regulatory standards.

RULE	LAYER	DIMENSION	TEST	SEVERITY	CHECKED	FAILED	PASS VS TARGET	PASS	TARGET	STATUS
B1	Buildings	Uniqueness	osm_id is unique	Error	91,212	0		**100%**	100%	Met
B2	Buildings	Geometry	Valid geometry	Error	91,212	1		**>99.99%**	100%	Below
B3	Buildings	Plausibility	Area ≥ 10 m²	Warning	91,212	25		**99.97%**	99%	Met
B4	Buildings	Plausibility	Area ≤ 50,000 m²	Review	91,212	3		**>99.99%**	—	Review
R1	Roads	Uniqueness	osm_id is unique	Error	18,403	0		**100%**	100%	Met
R2a	Roads	Completeness	Major roads: name or ref	Error	686	252		**63.3%**	95%	Below
R2b	Roads	Completeness	Tertiary: name or ref	Warning	705	518		**26.5%**	80%	Below
R3	Roads	Validity	oneway in F / T / B	Error	18,403	0		**100%**	100%	Met
R4	Roads	Validity	maxspeed 20–140 km/h	Error	278	0		**100%**	100%	Met
R5	Roads	Consistency	Tunnel → layer < 0	Warning	68	42		**38.2%**	90%	Below
R6	Roads	Consistency	Bridge → layer > 0	Warning	99	0		**100%**	90%	Met
R7	Roads	Consistency	Layer ≠ 0 → bridge/tunnel	Warning	137	12		**91.2%**	90%	Met
R8	Roads	Completeness	No named major neighbour	Review	252	103		**59.1%**	—	Review
P1	POIs	Uniqueness	osm_id is unique	Error	1,203	0		**100%**	100%	Met
P2	POIs	Completeness	Commercial POIs named	Warning	685	213		**68.9%**	80%	Below

Meets target · Below target · Review (no target) · Target

Compliance by dimension

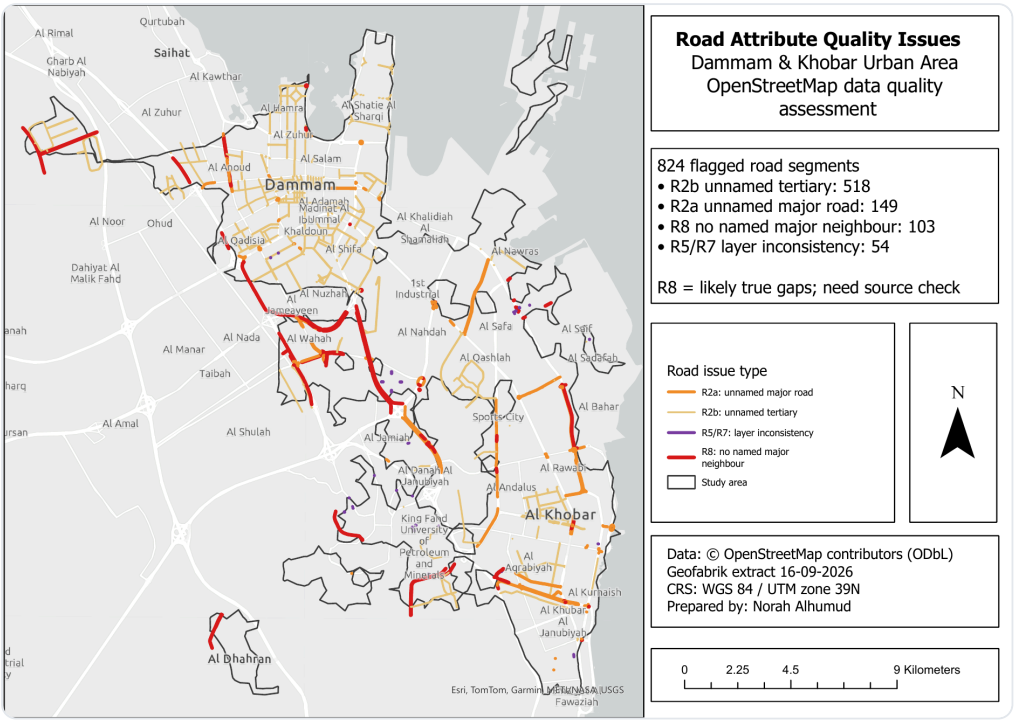
UNIQUENESS	GEOMETRY	PLAUSIBILITY	COMPLETENESS	VALIDITY	CONSISTENCY
3/3	**0/1**	**1/1**	**0/3**	**2/2**	**2/3**
targets met	targets met	targets met · 1 review	targets met · 1 review	targets met	targets met

Study area method. Building centroids inside the OSM admin boundaries of Dammam and Khobar were aggregated into polygons using ArcGIS Aggregate Points with a 500 m aggregation distance (UTM 39N). This produced 76 polygons (215.7 km²); the 7 polygons of 1 km² or more were kept as the study area (206.4 km²) and 69 smaller fragments (9.4 km²) were dropped. The 500 m and 1 km² thresholds are project choices and were not sensitivity-tested.

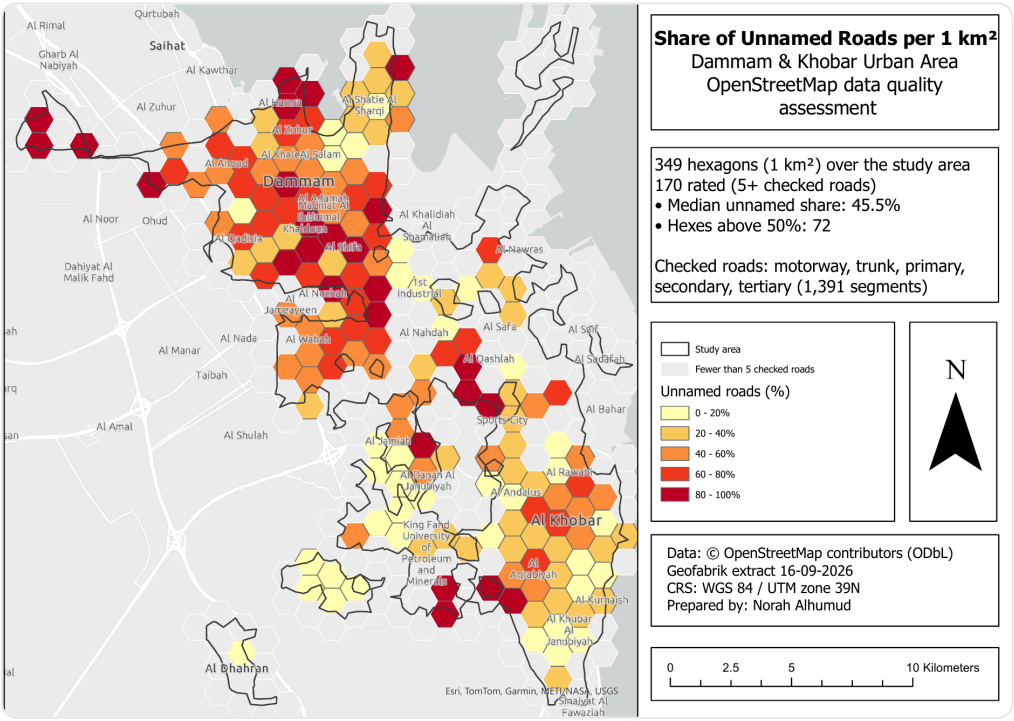
Empty values. Geofabrik stores missing text as a single space and unset numbers as 0. Every rule tests trimmed values, so a blank field counts as missing, not filled.

Where the gaps are

Central and northern Dammam carry most of the naming gap; the two maps show it by road and by area.



Road attribute issues. 824 flagged segments by type. Red marks R8: unnamed major roads that touch no named major road.



Unnamed roads per 1 km². 170 of 349 cells have five or more checked roads; their median unnamed share is 45.5%, and 72 cells exceed 50%.

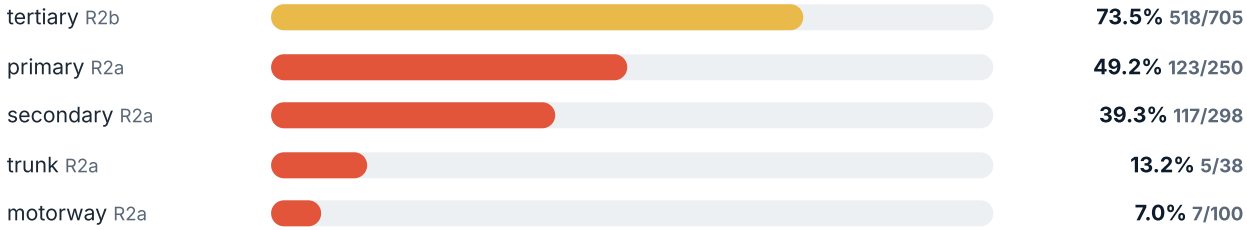
04 · THE NAMING GAP IN DETAIL

Which roads lack names

Tertiary roads have the highest share of missing names, but among Error-level classes primary and secondary roads carry most of the gap.

Missing names by road class

Share of segments with neither name nor ref. Red = Error-level classes (R2a), amber = tertiary (R2b).



Are the major-road gaps real?

The 252 unnamed major roads (R2a), split by whether they touch a named major road (R8).



103

Isolated: touch no named major road. Likely true gaps; verify first.

149

Attached to a named major road; may be ramps, links or split segments.

Not every flag is an error

Flags were sorted into classes before any correction is proposed.

CLASS	RULES	MEANING	FLAGS
Confirmed error	B2	Invalid building geometry	**1**
Likely true gap	R8	Unnamed major road touching no named major road	**103**
Probable continuation	R2a – R8	Unnamed major road touching a named major road	**149**
Tagging inconsistency	R5, R7	Tunnel or layer tags that disagree	**54**
Needs imagery check	B3, B4	Very small or very large building footprints	**28**
Optional attribute	R2b, P2	Unnamed tertiary road or commercial POI; OSM does not require a name	**731**

05 · ACTIONS

What to fix first

Effort goes to verified gaps and quick fixes first; optional attributes are tracked as improvement goals.

Recommendations

01 **Verify and name the 103 R8 roads first.** This alone lifts major-road naming from 63.3% to about 78%.

02 **Resolve the 149 attached segments** by confirming the parent road's name; together with step 1 this clears the 95% target.

03 **Start in central and northern Dammam,** where 1 km² cells most often exceed 60% unnamed roads.

04 **Fix the 54 tunnel and layer tags** and repair the invalid building; both are quick and improve routing and topology.

05 **Re-run the framework on each new extract** and track compliance in the dashboard release by release.

Limits. OSM was assessed against rules, not an official reference layer, so positional accuracy was not measured. Review flags (R8, B4) need source or imagery checks before they count as errors. Results describe one extract (16-09-2026).

Deliverables

Geodatabase
Raw layers kept intact, rules catalogue, results, violations and flagged layers

Quality maps
Road issues by type and unnamed-road share per 1 km²

Interactive dashboard
Compliance against targets and a 3D view of the naming gap

ArcPy notebook
Every check re-runnable on a new extract

PREPARED BY

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Data Quality & Analysis · Geospatial Information

ArcGIS Pro · ArcPy · OpenStreetMap
Geospatial Data Quality & Governance Framework · 2026

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